DEPERTMENT OF COMPUTER SCIENCE AND ENGINEERING

JATIYA KABI KAZI NAZRUL ISLAM UNIVERSITY, TRISHAL

Date: 25/05/2025

1. **Name of the student:** Md. Sohag Mia **Session:** 2023-24

   Roll Number: 24102040 **Reg. No.** 12996

   Name of the student: Tawhid Ahmed Abir **Session:** 2022-23

   Roll Number: 23102004 **Reg. No.** 11860

2. **Program:** BSc.

3. **Proposal for:** Project

4. **Name of the Proposed Supervisor: Professor Dr. Md. Sujan Ali**

   Designation: Professor

5. **Tentative Title:** Student Record Management System for CSE Department.

6. **Purpose of this project:** The purpose of this project is to design and implement a Student Record Management System (SRMS) tailored for the Department of Computer Science and Engineering. The system aims to efficiently manage and store student academic records, including personal information, course enrollment, grades, and performance reports, in a structured and user-friendly digital format.

7. Objectives with specific aims and future scope:

- ❖ To create a basic but effective record management system using the C programming language.
- ❖ To store, update, delete, and retrieve student data efficiently.
- ❖ To practice and apply programming fundamentals like file handling, structures, and data organization in C.
- ❖ To reduce manual workload and improve record accessibility for faculty and administrative staff.

8. Methodology:

The methodology involves the development of a command-line based student record system using the C programming language. The steps are as follows:

I. Requirements Analysis:

- ❖ Identify the types of data to be stored (e.g., student ID, name, department, CGPA).
- ❖ Determine the system's core functionalities (add, view, edit, delete records).

II. System Design:

- ❖ Design data structures using struct in C.
- ❖ Plan file storage using text/binary files for persistent data.

III. Implementation:

- ❖ Develop core modules:
  - o Add new student records.
  - o Search and view existing records.
  - o Update and delete records.
  - o Display all records.
- ❖ Implement file handling functions to store and retrieve data.

IV. Testing:

- ❖ Unit test each function individually.
- ❖ Perform integration testing for overall system performance.

V. Documentation:

- ❖ Document code logic, user instructions, and limitations.

9. Expected Results :

- ❖ A functioning Student Record Management System that can:
  - o Efficiently handle student data.
  - o Read and write student information from/to files.
  - o Provide a text-based interface for user interaction.
- ❖ Hands-on experience in implementing real-life applications using C.
- ❖ Practical understanding of file operations, and modular programming.

10. Experimental Design:

- ❖ **Analysis of required data fields.**
- ❖ **Development using a step-by-step approach.**
- ❖ **Implementation of file-based storage.**
- ❖ **Testing the system using sample data.**
- ❖ **Finalizing and validating the system.**

11. Future scope:

- ❖ Migration to a graphical or web-based interface.
- ❖ Addition of authentication and access control.
- ❖ Integration with university portals.
- ❖ Auto-email process for report cards

- ❖ Support for analytics and report generation for faculty.

12. Department's (Higher Study Committee) Or (Academic Committee Meeting) reference: (Filled up by HSC/Dept. Head)

Meeting No. **Resolution No.** **Date:**

Signature of Supervisor